

Bank Management System

Test Plan — Test Cases

(Deliverable 4: Test cases update to the Test Plan document, IEEE 829 format)

1. Purpose and Scope

This section supplements the Bank Management System Test Plan with a concrete set of test cases. It ensures a sufficient number of test cases exist covering both functional requirements (FR-1 to FR-14) and non-functional requirements (NFR-1 to NFR-9) as specified in the Software Requirements Specification (SRS), Version 1.0, dated 11/09/2026.

Seventeen test cases are defined below (TC-01 to TC-17), covering the core FRs across authentication, customer/account management, transactions, transfers, reporting and staff management, plus representative NFRs for performance, atomicity/safety, security and reliability. Section 5.1 security-validation cases are highlighted where applicable (TC-02, TC-06, TC-15, TC-16).

2. Requirement Traceability Summary

The table below provides traceability between the SRS requirement IDs and the test case(s) that validate them, in support of the Test Plan's traceability section.

Requirement ID	Requirement Summary	Covered By
FR-1	User login, RBAC enforcement	TC-01
FR-2	Logout, account lockout after 3 failed attempts	TC-02
FR-3	Customer record create / validation	TC-03, TC-04
FR-4	Account opening, minimum-balance check	TC-05
FR-5	Freeze / unfreeze / close account	TC-06
FR-6	Deposit transaction	TC-07
FR-7	Withdrawal, minimum-balance & limit checks	TC-08
FR-8	Fund transfer, atomicity	TC-09, TC-10
FR-9 / FR-10	Balance inquiry & transaction history, access scoping	TC-11
FR-11	Interest computation, no double-credit	TC-12
FR-13	Staff account deactivation, historical attribution	TC-13
NFR-1	Transaction performance ($\leq 2s$)	TC-14
NFR-3	Atomicity / no negative or sub-minimum balance	TC-09
NFR-4	Password/PIN hashing, no plain-text exposure	TC-15
NFR-5	Append-only audit trail	TC-16
NFR-7	Graceful handling of invalid input	TC-17

Note: FR-12 (statement export) and FR-14 (role dashboards) are covered by exploratory/UI verification during system testing rather than a dedicated scripted test case, and can be added as TC-18/TC-19 if the evaluator requires one test case per FR without exception.

3. Detailed Test Cases

FR-prefixed rows are functional test cases; NFR-prefixed rows are non-functional. Each row states preconditions and steps, the expected result, and a priority derived from the requirement's priority in the SRS.

TC ID	Req(s)	Objective	Preconditions / Steps	Expected Result	Priority
TC-01	FR-1	Verify a registered user can log in with valid credentials and is routed to the correct role dashboard. Type: Functional	Pre: User account exists and is active (role = Bank Officer). 1. Navigate to the login screen. 2. Enter a valid user ID and password/PIN. 3. Submit the login form.	User is authenticated and redirected to the Bank Officer dashboard; session is created.	High
TC-02	FR-1, FR-2	Verify the account is locked after three consecutive failed login attempts and access remains blocked until an Administrator unlocks it. Type: Functional	Pre: User account exists and is active. 1. Submit login with an incorrect password 3 times in a row. 2. Attempt to log in again with the correct password.	After the 3rd failed attempt the account is locked; step 2 is rejected with a message to contact an Administrator, even though the password is correct.	High
TC-03	FR-3	Verify a Bank Officer can create a new customer record with valid mandatory details. Type: Functional	Pre: Bank Officer is logged in. 1. Open 'New Customer' form. 2. Enter valid name, DOB, address, phone number and ID reference. 3. Submit the form.	Customer record is created with a unique customer identifier and is visible in the customer list.	High
TC-04	FR-3	Verify the system rejects a customer record submitted with a missing mandatory field. Type: Functional	Pre: Bank Officer is logged in. 1. Open 'New Customer' form. 2. Leave the phone number field empty; fill the rest. 3. Submit the form.	Submission is rejected; the phone number field is specifically flagged as invalid/missing. No record is created.	Medium
TC-05	FR-4	Verify an account cannot be opened with an initial deposit below the account type's minimum balance. Type: Functional	Pre: Customer is active; Savings account type has minimum balance = INR 1000. 1. Open 'New Account' form for the customer, select Savings. 2. Enter an initial deposit of INR 500. 3. Submit.	Request is rejected; system displays the required minimum balance (INR 1000). No account is created.	High
TC-06	FR-5	Verify a frozen account rejects deposits, withdrawals and transfers while remaining viewable.	Pre: Active account A1 exists with balance > 0; Administrator freezes A1. 1. Attempt a deposit to A1. 2. Attempt a withdrawal from A1.	Steps 1 and 2 are rejected with a 'frozen account' message and balance is unchanged; step 3 succeeds and shows correct balance/history.	High

TC ID	Req(s)	Objective	Preconditions / Steps	Expected Result	Priority
		Type: Functional	3. View A1's balance and history.		
TC-07	FR-6	Verify a deposit updates the account balance and records a corresponding transaction entry. Type: Functional	Pre: Active account A1, balance = INR 5000. 1. Deposit INR 2000 into A1. 2. Confirm the transaction. 3. View A1's transaction history.	Balance becomes INR 7000; a new 'Deposit' entry appears with correct amount, timestamp, resulting balance and operator ID.	High
TC-08	FR-7	Verify a withdrawal that would take the balance below the account's minimum balance is rejected and the balance is left unchanged. Type: Functional	Pre: Active Savings account A1, balance = INR 1500, minimum balance = INR 1000. 1. Attempt to withdraw INR 800 from A1. 2. Submit.	Transaction is rejected with a message stating the minimum-balance violation; A1's balance remains INR 1500.	High
TC-09	FR-8, NFR-3	Verify a fund transfer between two active accounts is applied atomically and generates two linked transaction entries. Type: Functional	Pre: Active accounts A1 (balance 5000) and A2 (balance 2000), different customers. 1. Transfer INR 1000 from A1 to A2. 2. Confirm the transaction. 3. Inspect transaction history of both accounts.	A1 = 4000, A2 = 3000; both accounts show a transaction entry with a shared transfer reference; if any step of the operation were forced to fail, neither balance would change.	High
TC-10	FR-8	Verify a transfer is rejected when the source and destination account are the same. Type: Functional	Pre: Active account A1 exists. 1. Initiate a transfer from A1 to A1. 2. Submit.	Transfer request is rejected before any balance is modified; an appropriate error message is shown.	Medium
TC-11	FR-9, FR-10	Verify a Customer can view balance and transaction history only for accounts they own. Type: Functional	Pre: Customer C1 owns account A1; account A2 belongs to another customer. 1. Log in as C1. 2. View balance/history of A1. 3. Attempt to view balance/history of A2 (e.g. by direct account lookup).	Step 2 succeeds and shows correct data in reverse-chronological order. Step 3 is denied and the access attempt is recorded.	High
TC-12	FR-11	Verify interest is not credited twice for the same account and period. Type: Functional	Pre: Interest-bearing account A1; interest run for Sept-2026 already completed. 1. Administrator re-triggers the interest run for Sept-2026.	Run is rejected/skipped for A1 with a 'already processed' message; no duplicate interest transaction is created.	Medium
TC-13	FR-13	Verify a deactivated Bank Officer account can no longer log in, while past transactions still show that officer's ID.	Pre: Bank Officer B1 has processed at least one past transaction. 1. Administrator deactivates B1. 2. Attempt to log in as B1.	Step 2 is rejected. Step 3 still shows B1 as the operator on the historical record.	Medium

TC ID	Req(s)	Objective	Preconditions / Steps	Expected Result	Priority
		Type: Functional	3. View a transaction previously processed by B1.		
TC-14	NFR-1	Verify a single balance-affecting transaction, including persistence, completes within 2 seconds under normal load. Type: Non-Functional (Performance)	Pre: Dataset of ~2000 accounts loaded; normal system load. 1. Perform a deposit transaction. 2. Measure elapsed time from submission to confirmation, including DB write.	Transaction completes and is persisted within 2 seconds.	Medium
TC-15	NFR-4	Verify passwords/PINs are never stored, logged or transmitted in plain text. Type: Non-Functional (Security)	Pre: A user account has been created with a known password. 1. Inspect the database table storing credentials. 2. Inspect application/server logs after a login attempt. 3. Capture network traffic during login (if HTTPS is disabled in a test environment).	The stored value is a salted hash, not the plain-text password; logs never contain the raw password; traffic capture shows no plain-text credential over HTTPS.	High
TC-16	NFR-5	Verify every balance-changing operation is captured in an append-only audit trail that cannot be edited or deleted via the application. Type: Non-Functional (Security / Audit)	Pre: At least one deposit and one transfer have been performed. 1. Open the audit trail as Administrator. 2. Attempt to edit or delete an existing audit entry through the application UI/API.	Audit trail lists timestamp, accounts affected, amount and operator for each entry; the edit/delete attempt in step 2 is refused by the application.	High
TC-17	NFR-7	Verify invalid input (non-numeric / negative amount, unknown account number) is handled gracefully without crashing or altering stored data. Type: Non-Functional (Reliability)	Pre: Active account A1, known balance. 1. Attempt a withdrawal with a negative amount. 2. Attempt a deposit with a non-numeric amount. 3. Attempt a transaction on a non-existent account number.	Each attempt is rejected with a clear error message; the application does not crash; A1's balance is unchanged after steps 1-2; no record is created for step 3.	Medium